Material Integrity Rules

Procurement Type

Fields used: Procurement Type, Plant Type, Material Type
Allowed domain: E, F, X (and **mandatory**)
Precedence:

1. **Plant Type takes precedence** (when present):
    - PP → Procurement must be F or X.
    - MP or FP → Procurement must be E or X.
2. If Plant Type does **not** decide, apply **Material Type** rule:
    - FERT/HALB → E or X.
    - ROH → F or X.

Account Assignment Group (MVKE)

Scope: Materials whose **Material Type ∈ {FERT, HALB, ROH}** (from SAPUI5).
External data: **MVKE.xlsx** is required.
Checks:

1. **Blank AAG**: Every MVKE row for the material must have **Acct Assmt Grp Mat. not empty** (each blank is an error).
2. **"Z1" coverage** (per material): If **any** MVKE row has AAG filled, at least **one** must be **Z1**; otherwise error "Missing 'Z1'".
3. **Coverage for FERT** across Sales Org/Channel:
    - If **PT00** exists with **DC 10 or 20**, then **US00** with **DC 30 must also exist**; if not, error.

Automatic Purchase Order Allowed

Field: Automatic Purchase Order Allowed, Plant Type
Rule:

- Mandatory (not blank).
- If Plant Type = PP → must be **Yes** (1).
- Otherwise → must be **No** (0)

Batch Management Required Ind.

Field: Batch Management Required Ind., Plant Type
Rule:

- Mandatory (not blank).
- If Plant Type ∈ {PP, COI} → must be **Yes** (1).
- Otherwise → must be **No** (0).

Consumption Mode

Field: Consumption Mode, Material Type
Rule:

- For **FERT**: value must be **0 or empty**; other values are invalid.
- For **non-FERT**: field is **mandatory** and must be **2**.

Division

Field: Division
Rule:

- Mandatory; must equal **00** (normalizing 0, 0.0 → 00).

Export/Import Group *(renamed to Intrastat Group in outputs)*

Field: Export/Import Group
Rule:

- Mandatory; must be **ZVIS**.
  (The reporting later renames the label to "Intrastat Group", but the validation uses the original field.)

General Item Category Group

Field: General Item Category Group
Rule:

- Mandatory; must be **NORM**.

Gross Weight (FP only)

Fields: Gross Weight, Net Weight, Unit of Weight (ISO Format), Plant Type
Scope: Only when Plant Type = FP.
Rules:

- Gross Weight and Net Weight **mandatory** (> 0), and **must be equal**.
- Upper bounds by **Unit**:
  - KG → ≤ **5**
  - G → ≤ **5000**
  - MG → ≤ **5,000,000**
  - Other units → ≤ **10**
- Non-numeric values trigger "invalid value" error.

Net Weight (FP only)

Fields: Net Weight, Gross Weight, Unit of Weight (ISO Format), Plant Type
Scope: Only when Plant Type = FP.
Rules (mirror of Gross):

- Gross & Net **mandatory**, **> 0**, and **equal**.
- Same upper bounds by Unit (KG 5; G 5000; MG 5,000,000; else 10).
- Non-numeric → error.

Indicator: Repetitive Manufacturing Allow

Fields: Indicator: Repetitive Manufa. Allow, Material Type, Plant Type
Rule:

- Mandatory.
- For **ROH**:
    - If Plant Type ∈ {MP, FP} → **Yes** (1);
    - If other Plant Type (not blank) → **No** (0).
- For **HALB**:
    - If Plant Type = PP → **must NOT** be Yes (1);
    - For other Plant Types (not blank) → **Yes** (1).
- For **FERT**:
    - If Plant Type = PP → **must NOT** be Yes (1);
    - For other Plant Types (not blank) → **Yes** (1).

Item Category Group (MVKE "ICG")

External data: **MVKE.xlsx**
Scope: Materials with **Material Type ∈ {FERT, HALB, ROH}** in SAPUI5.
Rule:

- If a material **exists in MVKE**, then **all** its MVKE rows must have **Item Category Group = NORM**.
- If the material doesn't exist in MVKE, it's **ignored** (no error).

JIT Delivery Sched.

Fields: JIT Delivery Sched., Plant Type
Rules:

- For Plant Type = PP: **mandatory** and must be **Yes** (1.0 as it appears).
- For **other** Plant Types: must be **empty**.

Loading Group

Field: Loading Group
Rule:

- Mandatory; must be **0001** (normalizes numeric forms to four-digit code).

Lot Sizing Procedure

Fields: Lot Sizing Procedure, Plant Type, Material
Rules:

- **Mandatory**.
- **Exception** for Plant Type = MP when **Material** contains token **14K450** (as a whole token): value can be **ZK or EX**.
- **Default rules** otherwise:
    - EX allowed **only** for Plant Type ∈ {FP, MP}.

- ZK allowed **only** for Plant Type = PP.

Material Group

Fields: Material Group, Material Type
Rules:

- Mandatory.
- Disallowed combinations:
  - HALB **cannot** be L004 or APPOINT.
  - FERT **cannot** be L003 or APPOINT.
  - ROH **cannot** be L003 or L004.

Period Type

Field: Period Type
Rules:

- Mandatory.
- Allowed values: T, P, M, W

Planning Calendar

Fields: Planning Calendar, Plant, Plant Type, Plant Specific Material Status, Material
PSMS gating: Validate **only** if PSMS in **[10..70]** (inclusive).
Overrides:

- If **Material contains 14K450 and** Plant Type = MP:
  - **Must not be empty**;

    - **Must not** be PT2.
      (This override short-circuits normal rules.)
      Normal rules:

- For Plant = PT01:
    - Plant Type ∈ {FP, MP} → Planning Calendar = PT1;
    - Plant Type = PP → must **not be empty** and **must not be PT2**.
- For Plant = PT02:
    - Plant Type ∈ {FP, MP} → Planning Calendar = PT2;
    - Plant Type = PP → must **not be empty**.

Planning Time Fence

Fields: Planning Time Fence, MRP Type, Plant Specific Material Status
PSMS gating:

- If PSMS = 10 → **skip** validation for that row.
  Value parsing: Non-numeric → **invalid**.
  Rules by MRP Type:
- PD → Planning Time Fence must be **empty or 0**.
- P1 or M1 → Planning Time Fence must be **> 0** (mandatory positive).

MRP Type *(with ZPLM MARC transfer awareness & PSMS precedence)*

Fields: MRP Type, Plant Type, Planning Time Fence, Plant, (optional) Plant Specific Material Status, (optional) Planning Calendar, Material
Aux data: Transfer flag from **ZPLM MARC** (Material Transfer) per (Material, Plant).
Permitted domain: M1, PD, P1, ND, V1.
PSMS precedence:

- PSMS = 10 → **skip** row.
- PSMS = 25 → **forced**:

    - MRP Type = ND
    - Planning Time Fence = 0 or empty
    - If Plant = PT01 → Planning Calendar = PT1; if Plant = PT02 → PT2.
- PSMS ∈ {70, 75} → **forced**:
    - MRP Type = ND (other checks not applied).
      Base rules (when not PSMS 10/25/70/75):
- **Mandatory** & in allowed set.
- **FP (not transfer)**: MRP = M1 and **Time Fence > 0**.
- **FP transfer or MP transfer**: MRP ∈ {M1, P1, PD}; Time Fence free (no constraint).
- **MP (no transfer)**: MRP = PD and **Time Fence = 0 or empty**.
- **PP**:
    - If **Time Fence > 0** → MRP = P1;
    - If **Time Fence = 0/empty** → MRP = PD.
      (Invalid/unknown numeric in Time Fence → error.)

Price Control

Field: Price Control
Rules:

- Mandatory; must be **S**.

Product Hierarchy (SAPUI5)

Field: Product Hierarchy, Material Type
Rule:

- For **FERT only** → **mandatory** (not blank).

Product Hierarchy vs Enterproj Number (ZPLM MARA cross-check)

Data: MARA (Product Hierarchy) vs **ZPLM MARA** (Enterproj Number).
Scope: **FERT** only.
Rule:

- Validate that the substring **after the P at the 13th position** in **Product Hierarchy** equals **Enterproj Number** (after normalization: numeric trimming, zero-pad to PH suffix length, treat blanks/NaN as zeros).
- If no P at 13th position → **invalid PH**.
- If normalized suffix ≠ normalized Enterproj → **PH mismatch** error.

Production Scheduling Profile

Field: Production Scheduling Profile, Material Type
Rule:

- For **FERT** and **HALB**: **mandatory** and must be **ZREM01**.

Profit Center

Fields: Profit Center, Plant
Rules:

- Mandatory.
- For Plant = PT01 → Profit Center = PT00351.
- For Plant = PT02 → Profit Center = PT00361.

Purchasing Group

Fields: Purchasing Group, Plant Type
Rule:

- **Mandatory only when Plant Type = PP** (blank allowed otherwise).

Repetitive Manufacturing Profile

Fields: Repetitive Manufacturing Profile, Material Type
Rules:

- For **FERT** and **HALB**: **mandatory** and must be **Z002**.
- For **ROH**: **must not** be Z002.

Source List Required

Fields: Source List Required, Plant Type
Rules:

- Mandatory.
- If Plant Type = PP → **Yes** (1)
- Otherwise → **No** (0).

Issue Storage Location

Fields: Issue Storage Location, Plant
Rules:

- Mandatory.
- P105 → **only** allowed for Plant = PT01.
- P205 or P220 → **only** for Plant = PT02.

Valuation Class

Fields: Valuation Class, Material Type
Rules:

- Mandatory.
- 3000 → **only** for ROH.
- 7900 → **only** for HALB.
- 7920 → **only** for FERT.

With Qty Structure

Field: With Qty Structure
Rule:

- **Mandatory**; must be **1** ("Yes").

Enterproj Number (in SAPUI5 Export)

Fields: Enterproj Number, Material Type, Material
Rules:

- For **FERT** → **mandatory** (not blank).
- For **HALB**/**ROH** → **must be empty**.
  (When present as numeric with .0, it is normalized only for the error message display.)

Inspection Type (QMAT cross-checks)

External data: **QMAT.xlsx**
Joins: MARC (by Material + Plant) to QMAT.
Rules:

- **Type 08**: For **HALB/ROH** materials, if QMAT has Inspection Type 08 for the (Material, Plant), then **Active flag must NOT be X** (active).
- **Type 1708**: For **FERT/HALB/ROH** materials, if QMAT has Inspection Type 1708, the **Active flag must NOT be X**.

Backflush

Field: Backflush
Rule: **Mandatory** (must not be empty).

Strategy Group

Field: Strategy Group
Rules:

- **Mandatory**.
- Must be **40** (normalizes numeric variants like 40.0).

Product-Related Origin

Field: Product-Related Origin
Rule: **Mandatory** (must not be empty)

Availability Check

Field: Availability Check
Rule: **Mandatory** (must not be empty).

MRP Controller

Field: MRP Controller
Rule: **Mandatory** (must not be empty).

Plant Specific Material Status *(PSMS)*

Field: Plant Specific Material Status
Rule: **Mandatory** (must not be empty).
*(Also used as a **gating/precedence** field in other validations such as **MRP Type, Planning Calendar**, and **Description** exception.)*

Negative Stocks in Plant

From validar_neg_stocks_in_plant.

Applies to **MARC** records.

Scope

Only for materials with Material Type:

- **FERT**
- **HALB**
- **ROH**

Rules

ROH / HALB

- Field **must NOT be empty** → mandatory

- If marked (X/1/TRUE/YES/Y), then:

Allowed only for Plant ∈ {PT01 with Location P105, PT02 with Location P205 or PT02 with status 25 or 75 and Location P220}

AND Material Type ∈ {HALB, ROH}

FERT

- Negative stock flags are **ignored** (no errors).

Extra (engine-side enrichments that affect reporting—but not strictly SAPUI5 validations)

- **"Old material number"** (MARA): for materials present in prior report errors, it must **equal the Material without hyphens** and must not be blank—reported in "All Errors" as an additional cross-file check

---

2. FP Version Cleanup Rules

These rules apply to **materials where Material Plant Type = 'FP'** in **ZPLM MARC**.

2.1 Version Families

Materials are grouped by:

Family = Material without the last character

Example:

- ABC123A → Base: ABC123
- ABC123B → Base: ABC123

The engine **keeps only families where Base appears ≥ 2 times**.

(Implemented via _filter_materials_by_base_duplicates.)

2.2 Classification Within Each Family

Inside each group (Plant, Base):

- Versions are sorted by the **final character** (A–Z).
- The **highest alphabetical version is "Correct"**, all others become **"Outdated"**.

(Implemented in _classify_fp_versions.)

---

3. FP Status Rules (PSMS Rules)

Applied in run_fp_status_validation.

For materials where:

- Material Plant Type = **FP** (from ZPLM MARC)
- PSMS ∈ **{40, 45, 50, 55}** (from MARC)

3.1 Allowed PSMS Values

Only **40, 45, 50, 55** are valid for FP cleanups.

3.2 PSMS Naming

Mapped internally for reporting:

PSMS	Meaning
40	NewProd
45	Prod
50	Balout
55	Service

Used only for reporting.

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4. FP Rule — Status "55" Cannot Coexist in PT01 and PT02

From validar_fp_pt01_pt02_status55.

Rule

If a material exists as **FP** in *both*:

- Plant **PT01**
- Plant **PT02**

Then **PSMS must NOT be 55** in MARC for *either* plant.

Error

“Material with FP in PT01 AND PT02 cannot have status 55”

---

5. FP Rule — Service Liability vs EOP Date (Corrected Rule)

From validar_fp55_eop_vs_service_liability.

Applies to:

- Material Plant Type = **FP**
- PSMS = **55**
- On a (Material, Plant) level
- Using ZPLM MARC for EOP/Service Liability dates

5.1 Required Fields

- **Service Liability Date must NOT be empty**
  If empty → *Error*

5.2 Chronological Rule

The **EOP Date must be STRICTLY EARLIER than Service Liability Date**.

If:

EOP ≥ Service Liability → ERROR

This is the corrected final business rule.

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6. FP Rule — Expired Service Liability Date

From validar_fp55_service_liability_expired.

Applies to:

- FP Materials
- PSMS = **55**

Rule

If Service Liability Date is **≤ Today**, then:

ERROR: “Service Liability Date is already expired (≤ today)”

Works only if SL date is filled.

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7. Standard Price Required for FP/MP/PP with PSMS=70

From validar_fppp70_standard_price_mbew.

Applies to materials where:

- Material Plant Type ∈ **{FP, MP, PP}** (from ZPLM MARC)
- PSMS = **70** (from MARC)
- Standard Price comes from **MBEW** (STPRS)

Rule

- Standard Price is **mandatory**
- Standard Price must be **> 0**

If empty or zero:

ERROR: “Standard Price is empty or zero for FP/MP/PP with status 70”

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8. MARA/MARC/ZPLM Cross-File Consistency Checks

These rules ensure structural data integrity.

8.1 MARA → ZPLM MARA Existence

From validar_mara_vs_zplm_mara_existence.

For FERT/HALB/ROH:

If material exists in MARA but **missing from ZPLM MARA**:

ERROR: “Missing in ZPLM MARA”

8.2 ZPLM MARA → MARA Existence

From validar_zplm_mara_vs_mara_existence.

For FERT/HALB/ROH:

If material exists in ZPLM MARA but not in MARA (filtered set):

ERROR: “Exists in ZPLM MARA but missing in filtered MARA”

8.3 MARC → ZPLM MARC Existence

From validar_marc_vs_zplm_marc_existence.

If (Material, Plant) in MARC but *missing* in ZPLM MARC:

ERROR: “Missing in ZPLM MARC”

8.4 ZPLM MARC → MARC Existence

From validar_zplm_marc_vs_marc_existence.

If (Material, Plant) in ZPLM MARC but not present in MARC:

ERROR: “Exists in ZPLM MARC but missing in MARC”